

M. Shahbaz Murtaza

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EDUCATION

University of British Columbia

Bachelor of Science in Statistics

Vancouver, BC

High Distinction; International Major Entrance Scholar

EXPERIENCE

Risk Management Intern

June 2025 – Aug. 2025

Soneri Bank

Hybrid

- Performed internal ratings for clients by analyzing financial statements, cash flows and qualitative metrics.
- Validated and cleaned insurance credit risk model inputs (PD, LGD) in Excel using nested formulas, pivot tables, scenario analysis, and data validation rules, improving data accuracy and model reliability by 15%.
- Translated complex risk concepts into clear dashboards and written summaries for non-technical stakeholders.
- Proposed automation of financial modeling workflows using SQL-based data warehousing and Python pipelines.

Vice President, External

Sept. 2024 – May 2025

UBC Actuarial Science Club

Vancouver, BC

- Curated interactive social media campaigns to boost visibility, resulting in 40K+ views over the last five months.
- Secured partnerships with organizations while maintaining relationships with industry experts for future events.
- Resolved conflicts, supervised club events and delegated tasks accordingly to maximize student engagement.
- Collaborated with the team to oversee recruitment; conducted interviews from the selected set of 43 applicants.

Graphic Designer

Sept. 2025 – present

Get Thrifty UBC

Vancouver, BC

- Designed data-informed visuals by analyzing engagement metrics, improving audience interaction by 50%.
- Optimized visual layouts using A/B testing principles to improve clarity and user response.
- Applied UX-inspired design principles to enhance readability and user retention.

Auxiliary Sales Attendant

Sept. 2022 – Dec. 2025

Tim Hortons

Vancouver, BC

- Provided exceptional customer service to all visitors and increased customer satisfaction survey scores by 20%.
- Managed multiple concurrent tasks under time constraints, prioritizing deliverables in a fast-paced environment.

PROJECTS

Bayesian Volatility Modeling of Tech Stocks | R, Stan, Git

Jan. 2025 – Apr. 2025

- Modeled daily returns of META, AAPL, AMZN, NFLX, GOOGL, and XLK ETF using a Bayesian hierarchical AR(1)-GARCH(1,1) framework with Student-t innovations to capture volatility clustering and fat tails.
- Implemented model in using Hamiltonian Monte Carlo with 3 chains $\times$ 2000 iterations, achieving convergence ($\hat{R} < 1.05$) and ESS > 300 for key parameters.
- Identified sector-level volatility persistence ($\beta \approx 0.899$) and autocorrelation ($\phi \approx 0.117$), revealing differential risk profiles across firms during market shocks (e.g., COVID-19, 2022 crash, 2025 US Tariff Impositions.)
- Enhanced robustness to outliers and tail risk, enabling improved volatility forecasts and risk estimation.

Fitness Database | JavaScript, SQL, Git

Jan. 2025 – Apr. 2025

- Built a normalized SQL database (3NF/BCNF) for a fitness center, modeling entities like users, workouts, trainers, and diets with enforced referential integrity.
- Developed a front-end GUI using JavaScript, enabling dynamic interaction with the backend for executing 10+ analytical queries (e.g., caloric intake, trainer assignments, user activity).
- Collaborated in a team of 3 to design and integrate database and front-end layers, deploying a functional system aligned with real-world business logic.

Optimizing Diabetes Prediction | R, Git

Oct. 2024 – Dec. 2024

- Developed cross-validated, feature-engineered models using Logistic Regression and LASSO regularization.
- Visualized scatterplot and correlation matrices, histograms and AUC-ROC curves for deeper inspection.
- Leveraged sensitivity analysis to assist comparative analysis between the models, achieving a 77% accuracy.

MISCELLANEOUS

Languages/Frameworks: MS Excel, MS Word, R, SQL, Python, Figma, Javascript, Canva, Stan, Power BI

Certifications: EXAM P (ASA), Financial Markets (Yale University), EXAM FM (sitting)